Signal Engine: Mathematical Specification

Unicorn Hunt — Small-Cap Momentum Screener

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1 Overview and Notation

Let $\mathcal{U} = \{s_1, s_2, \ldots, s_N\}$ denote the investable universe of N tickers. For each ticker $s \in \mathcal{U}$ and trading day t, we observe the standard OHLCV tuple $(O_t^s,\ H_t^s,\ L_t^s,\ C_t^s,\ V_t^s)$. Days are indexed backward from the most recent close: $t = 0$ is today, $t = -1$ is the previous trading day, and so on.

The screener evaluates each ticker across $K = 8$ signals. Each signal k proceeds in two stages:

1. **Raw scoring.** A function $f_k \colon \mathcal{U} \to \mathbb{R}$ maps each ticker to an unbounded raw value, where positive values are bullish and negative values are bearish.
2. **Percentile normalisation.** Raw values are converted to a $[0, 100]$ percentile score $\sigma_k(s)$ with the convention that 50 is neutral.

1.1 Percentile Normalisation

Let $\{r_1, r_2, \ldots, r_N\}$ be the raw signal values for all N tickers. The population is split at zero:

$$\mathcal{P} = \{r_i : r_i > 0\}, \qquad \mathcal{N} = \{r_i : r_i < 0\}, \qquad \mathcal{Z} = \{r_i : r_i = 0\} \tag{1}$$

For a ticker with raw value r:

$$\sigma(r) = \begin{cases} 50 + 50 \cdot \text{percentile}(r \mid \mathcal{P})/100 & \text{if } r > 0, \\ 50 - 50 \cdot \big(1 - \text{percentile}(r \mid \mathcal{N})/100\big) & \text{if } r < 0, \\ 50 & \text{if } r = 0, \end{cases} \tag{2}$$

where $\text{percentile}(r \mid S)$ is the percentile rank of r within set S, scaled $[0, 100]$. This ensures the neutral midpoint at 50 is preserved regardless of the distribution's skew.

1.2 Composite Score

Given signal weights $w_k \geq 0$ with $\sum_{k=1}^{K} w_k = 1$, the composite score for ticker s is:

$$\Sigma(s) = \sum_{k=1}^{K} w_k \, \sigma_k(s) \tag{3}$$

If a ticker lacks data for signal k, that signal defaults to the neutral value $\sigma_k(s) = 50$.

1.3 Default Weight Vector

Signal	Weight w_k
Price Momentum	0.20
Volume Surge	0.20
Insider Activity	0.20
Financial Health	0.15
Stochastic	0.10
Price Acceleration	0.10
News Attention	0.05
RSI	0.00

2 Price Momentum

2.1 Motivation

Classical cross-sectional momentum (Jegadeesh & Titman, 1993) establishes that stocks with strong intermediate-horizon returns tend to continue outperforming. The standard academic approach skips the most recent month to avoid the well-documented short-term reversal effect. This signal implements that approach with a multi-period blend.

2.2 Definitions

Let $\delta = 21$ trading days denote the skip period (approximately one calendar month). Define the n-day lagged return, measured from n days before the skip point to the skip point:

$$R_n(s) = \frac{C^s_{-\delta} - C^s_{-(\delta+n)}}{C^s_{-(\delta+n)}} \tag{4}$$

Three lookback horizons are used:

Period	Lookback n (days)	Approx. calendar	Weight α
3-month	63	3 months	0.40
6-month	126	6 months	0.30
12-month	252	12 months	0.30

2.3 Raw Signal

$$f_{\text{mom}}(s) = 0.40\, R_{63}(s) + 0.30\, R_{126}(s) + 0.30\, R_{252}(s) \tag{5}$$

A ticker is excluded if fewer than $252 + \delta = 273$ trading days of history are available, or if any denominator price is non-positive.

3 Volume Surge

3.1 Motivation

Elevated trading volume on a directional move signals conviction. A breakout on three times normal volume is far more significant than one on average volume. By normalising volume against the stock's own history *and* its market capitalisation, the signal captures genuine accumulation or distribution.

3.2 Definitions

Let $\bar{V}_n^s$ denote the simple average daily volume over the most recent n days:

$$\bar{V}_n^s = \frac{1}{n} \sum_{i=0}^{n-1} V_{-i}^s \tag{6}$$

Define the **volume ratio** (short window $m = 5$ days, long window $n = 60$ days):

$$\rho(s) = \frac{\bar{V}_5^s}{\bar{V}_{60}^s} \tag{7}$$

Define the **dollar turnover** relative to market capitalisation M^s:

$$\tau(s) = \frac{\sum_{i=0}^{4} C_{-i}^s \cdot V_{-i}^s}{M^s} \tag{8}$$

Define the **5-day price return**:

$$r_5(s) = \frac{C_0^s - C_{-4}^s}{C_{-4}^s} \tag{9}$$

The **direction indicator** is:

$$d(s) = \begin{cases} +1 & \text{if } r_5(s) > 0 \\ -1 & \text{if } r_5(s) < 0 \\ +0.5 & \text{if } r_5(s) = 0 \end{cases} \tag{10}$$

The flat-price case receives a mild positive bias: elevated volume with stable price suggests quiet accumulation.

3.3 Raw Signal

$$f_{\text{vol}}(s) = \rho(s) \cdot \tau(s) \cdot d(s) \tag{11}$$

A ticker is excluded if fewer than 60 days of history are available, if average historical volume is non-positive, or if market capitalisation is unavailable.

4 Insider Activity

4.1 Motivation

Corporate insiders—officers, directors, and significant shareholders—must disclose transactions via SEC Form 4 filings. In the small-cap universe, insider purchases are among the strongest known predictive signals: insiders commit personal capital and possess material non-public knowledge of the business. Insider selling is a weaker signal in the opposite direction because insiders sell for many non-informative reasons (diversification, tax obligations, personal liquidity).

4.2 Definitions

For each ticker s, the following quantities are observed over a 90-day lookback window:

- $n_f(s)$ — total number of Form 4 filings
- $n_b(s)$ — number of purchase transactions
- $n_s(s)$ — number of sale transactions
- $B(s)$ — total dollar value of purchases
- $S(s)$ — total dollar value of sales
- $\Delta(s) = B(s) - S(s)$ — net buy value

Let $\tilde{B}$ and $\tilde{S}$ denote the universe-wide median purchase value and median sale value respectively, computed only over tickers with non-zero values.

4.3 Component 1: Direction Score (weight 0.60)

$$D(s) = \begin{cases} \min\left(\dfrac{B(s)}{\tilde{B}},\ 5\right) \Big/ 5 & \text{if } \Delta(s) > 0 \\ -\dfrac{1}{2}\ \min\left(\dfrac{S(s)}{\tilde{S}},\ 5\right) \Big/ 5 & \text{if } \Delta(s) < 0 \\ 0 & \text{if } \Delta(s) = 0 \end{cases} \tag{12}$$

When $\Delta(s) > 0$ and $n_s(s) = 0$ (pure buying with no sales), the direction score receives a conviction bonus:

$$D^*(s) = \min\big(1.3 \cdot D(s),\ 1.0\big) \tag{13}$$

The factor of $\frac{1}{2}$ in the selling branch encodes the intentional asymmetry: selling is penalised at half the rate that buying is rewarded.

4.4 Component 2: Activity Level (weight 0.40)

$$A(s) = \min\left(\frac{n_f(s)}{10},\ 1.0\right) \tag{14}$$

Activity amplifies the direction signal—it does not contribute independently.

4.5 Raw Signal

$$f_{\text{insider}}(s) = \text{clip}\big(0.6\, D^*(s) + 0.4\, A(s) \cdot D^*(s),\ \ -1,\ 1\big) \tag{15}$$

which simplifies to:

$$f_{\text{insider}}(s) = \text{clip}\big(D^*(s)\ \big[0.6 + 0.4\, A(s)\big],\ \ -1,\ 1\big) \tag{16}$$

Tickers with $n_f(s) = 0$ receive a raw signal of exactly zero (neutral).

5 Financial Health

5.1 Motivation

Momentum applied naïvely can lead investors into financially distressed companies experiencing short-lived speculative interest. This signal acts as a quality filter, scoring each ticker on solvency, cash reserves, profitability, and data recency.

5.2 Sub-Score Definitions

Each sub-score maps a continuous financial metric to a discrete $[0, 1]$ grade via piecewise-constant thresholds.

5.2.1 Solvency Score

Let CR denote the current ratio (current assets / current liabilities) and DE denote the debt-to-equity ratio (long-term debt / total equity).

$$g_{\mathrm{CR}} = \begin{cases} 1.00 & \text{if CR} \geq 2.0 \\ 0.75 & \text{if } 1.5 \leq \text{CR} < 2.0 \\ 0.50 & \text{if } 1.0 \leq \text{CR} < 1.5 \\ 0.25 & \text{if } 0.5 \leq \text{CR} < 1.0 \\ 0.00 & \text{if CR} < 0.5 \end{cases} \qquad g_{\mathrm{DE}} = \begin{cases} 1.00 & \text{if } 0 \leq \text{DE} \leq 0.5 \\ 0.75 & \text{if } 0.5 < \text{DE} \leq 1.0 \\ 0.50 & \text{if } 1.0 < \text{DE} \leq 2.0 \\ 0.25 & \text{if } 2.0 < \text{DE} \leq 4.0 \\ 0.00 & \text{if DE} > 4.0 \text{ or DE} < 0 \end{cases} \tag{17}$$

Negative debt-to-equity indicates negative shareholder equity and receives the minimum score. Missing values default to 0.50.

$$\text{Solvency}(s) = 0.6\, g_{\mathrm{CR}}(s) + 0.4\, g_{\mathrm{DE}}(s) \tag{18}$$

5.2.2 Cash Position Score

Let CA denote cash and equivalents as a fraction of total assets.

$$\text{Cash}(s) = \begin{cases} 1.00 & \text{if CA} \geq 0.30 \\ 0.75 & \text{if } 0.15 \leq \text{CA} < 0.30 \\ 0.50 & \text{if } 0.05 \leq \text{CA} < 0.15 \\ 0.25 & \text{if } 0.01 \leq \text{CA} < 0.05 \\ 0.10 & \text{if CA} < 0.01 \end{cases} \tag{19}$$

5.2.3 Profitability Score

Let μ denote the net profit margin (net income / revenue).

$$\text{Profit}(s) = \begin{cases} 1.00 & \text{if } \mu \geq 0.15 \\ 0.80 & \text{if } 0.05 \leq \mu < 0.15 \\ 0.60 & \text{if } 0.00 \leq \mu < 0.05 \\ 0.30 & \text{if } -0.10 \leq \mu < 0.00 \\ 0.15 & \text{if } -0.30 \leq \mu < -0.10 \\ 0.00 & \text{if } \mu < -0.30 \end{cases} \tag{20}$$

5.2.4 Filing Recency Score

Let Δd denote the number of calendar days since the most recent SEC filing.

$$\text{Recency}(s) = \begin{cases} 1.00 & \text{if } \Delta d \leq 120 \\ 0.75 & \text{if } 120 < \Delta d \leq 200 \\ 0.50 & \text{if } 200 < \Delta d \leq 400 \\ 0.25 & \text{if } 400 < \Delta d \leq 600 \\ 0.10 & \text{if } \Delta d > 600 \end{cases} \tag{21}$$

Unknown or unparseable filing dates receive 0.25.

5.3 Composite and Raw Signal

$$Q(s) = 0.35\ \text{Solvency}(s) + 0.25\ \text{Cash}(s) + 0.25\ \text{Profit}(s) + 0.15\ \text{Recency}(s) \tag{22}$$

The composite $Q(s) \in [0, 1]$ is centred and rescaled to a $[-1, +1]$ raw signal:

$$f_{\text{health}}(s) = 2\big(Q(s) - 0.5\big) \tag{23}$$

6 Stochastic Oscillator

6.1 Motivation

The stochastic oscillator (Lane, 1984) answers a simple question: *where did the stock close relative to its recent trading range?* If a stock has traded between \$10 and \$15 over the past two weeks and today it closed at \$14.50, it is closing near the top of its range — a sign of persistent buying pressure.

The raw oscillator is noisy, so two rounds of smoothing are applied, producing what is known as the "slow stochastic." This signal then extracts three momentum-relevant properties from the smoothed oscillator and combines them into a single score.

6.2 Input Data

For each ticker s and each trading day t, we observe four prices:

- O_t^s — the **opening** price on day t
- H_t^s — the **highest** price reached during day t
- L_t^s — the **lowest** price reached during day t
- C_t^s — the **closing** price on day t

Days are indexed backward from today: $t = 0$ is the most recent close, $t = -1$ is the previous trading day, $t = -2$ is two days ago, and so on.

Three fixed parameters govern the oscillator:

Parameter	Symbol	Value
Lookback period (range window)	p	14 days
Slow smoothing window	q	3 days
Signal line smoothing window	d	3 days

6.3 Step 1: Highest High and Lowest Low

For any given day t, look back over the preceding $p = 14$ trading days (including day t itself) and find the highest high and lowest low across that window:

$$\mathrm{HH}_t = \max\bigl(H_t^s,\ H_{t-1}^s,\ H_{t-2}^s,\ \ldots,\ H_{t-(p-1)}^s\bigr) \tag{24}$$

$$\mathrm{LL}_t = \min\bigl(L_t^s,\ L_{t-1}^s,\ L_{t-2}^s,\ \ldots,\ L_{t-(p-1)}^s\bigr) \tag{25}$$

In words: HH_t is the single highest price the stock touched on any day in the 14-day window ending at day t, and LL_t is the single lowest price the stock touched on any day in the same window. Together they define the stock's recent **trading range**.

6.4 Step 2: Fast %K (the raw oscillator)

The fast %K measures where today's close sits within the 14-day trading range, expressed as a percentage from 0 to 100:

$$\%K_t^{\text{fast}} = \frac{C_t^s - \text{LL}_t}{\text{HH}_t - \text{LL}_t} \times 100 \tag{26}$$

Interpretation:

- If $C_t^s = \text{HH}_t$ (the stock closed at its 14-day high), then $\%K_t^{\text{fast}} = 100$.
- If $C_t^s = \text{LL}_t$ (the stock closed at its 14-day low), then $\%K_t^{\text{fast}} = 0$.
- If the close is exactly halfway between the high and the low, $\%K_t^{\text{fast}} = 50$.

If $\text{HH}_t = \text{LL}_t$ (the stock traded at a single price for 14 days, i.e. the range is zero), the value is undefined and the ticker is excluded from scoring on this day.

Note that $\%K_t^{\text{fast}}$ is computed for *every* day t in the history, not just today. This produces a time series of values, one per day.

6.5 Step 3: Slow %K (first smoothing)

The fast %K is noisy because it responds to a single day's close. The **slow %K** smooths it by taking a simple moving average over $q = 3$ consecutive days:

$$\%K_t = \frac{1}{q}\sum_{i=0}^{q-1} \%K_{t-i}^{\text{fast}} = \frac{\%K_t^{\text{fast}} + \%K_{t-1}^{\text{fast}} + \%K_{t-2}^{\text{fast}}}{3} \tag{27}$$

This is the primary line of the slow stochastic oscillator. Like the fast version, it ranges from 0 to 100, but moves more gradually.

6.6 Step 4: Slow %D (second smoothing — the signal line)

The **slow %D** applies a second round of smoothing to the slow %K, again using a $d = 3$-day simple moving average:

$$\%D_t = \frac{1}{d}\sum_{i=0}^{d-1} \%K_{t-i} = \frac{\%K_t + \%K_{t-1} + \%K_{t-2}}{3} \tag{28}$$

The %D line is called the **signal line** because crossovers between %K and %D are traditionally used as buy/sell triggers. When %K crosses above %D, it suggests accelerating upward momentum; when %K crosses below %D, it suggests weakening momentum.

6.7 Step 5: Extract Three Momentum Components

With the slow stochastic computed for the full price history, we now read off three values from the most recent days and convert each into a normalised component score in the range $[-1, +1]$.

6.7.1 Component A: Level (weight 0.50)

Question: Is the stock closing near the top or the bottom of its recent range?

Take the current slow %K value (at $t = 0$) and measure its distance from the midpoint of 50, normalised so the full $[0, 100]$ range maps to $[-1, +1]$:

$$\ell(s) = \frac{\%K_0 - 50}{50} \tag{29}$$

Examples:

- $\%K_0 = 80$ (closing near the top of the range): $\ell = (80 - 50)/50 = +0.60$
- $\%K_0 = 50$ (closing at the midpoint): $\ell = 0.00$
- $\%K_0 = 20$ (closing near the bottom): $\ell = (20 - 50)/50 = -0.60$

This component receives the highest weight (50%) because the absolute position within the range is the most direct measure of momentum.

6.7.2 Component B: Crossover (weight 0.30)

Question: Is the %K line above or below its signal line %D, and by how much?

Take the spread between the current slow %K and slow %D, normalise by a reference spread of 20 points (chosen as a typical magnitude for meaningful crossovers), and clamp to $[-1, +1]$:

$$\chi(s) = \text{clip}\left(\frac{\%K_0 - \%D_0}{20},\ -1,\ +1\right) \tag{30}$$

Examples:

- $\%K_0 = 75,\ \%D_0 = 65$ (bullish crossover, %K above %D by 10 points): $\chi = 10/20 = +0.50$
- $\%K_0 = 40,\ \%D_0 = 55$ (bearish crossover, %K below %D by 15 points): $\chi = -15/20 = -0.75$
- $\%K_0 = \%D_0$ (lines are converged): $\chi = 0.00$

6.7.3 Component C: Trend (weight 0.20)

Question: Is the slow %K rising or falling right now?

Take the one-day change in slow %K (comparing today's value to yesterday's), normalise by a reference daily change of 10 points, and clamp:

$$\theta(s) = \text{clip}\left(\frac{\%K_0 - \%K_{-1}}{10},\ -1,\ +1\right) \tag{31}$$

Examples:

- $\%K_0 = 72,\ \%K_{-1} = 65$ (rising by 7 points): $\theta = 7/10 = +0.70$
- $\%K_0 = 45,\ \%K_{-1} = 50$ (falling by 5 points): $\theta = -5/10 = -0.50$

6.8 Step 6: Raw Signal (weighted combination)

The three components are combined using fixed weights that reflect their relative importance:

$$f_{\text{stoch}}(s) = 0.50\ \ell(s) + 0.30\ \chi(s) + 0.20\ \theta(s) \tag{32}$$

Worked example. Suppose a stock has $\%K_0 = 75$, $\%D_0 = 68$, $\%K_{-1} = 70$:

$$\ell = (75 - 50)/50 = +0.50 \tag{33}$$
$$\chi = \text{clip}\big((75 - 68)/20,\ -1,\ 1\big) = +0.35 \tag{34}$$
$$\theta = \text{clip}\big((75 - 70)/10,\ -1,\ 1\big) = +0.50 \tag{35}$$
$$f_{\text{stoch}} = 0.50 \times 0.50 + 0.30 \times 0.35 + 0.20 \times 0.50 = 0.455 \tag{36}$$

This positive value indicates bullish momentum: the stock is closing in the upper half of its range, %K is above its signal line, and the oscillator is rising. This raw value is then converted to a 0–100 percentile score via the normalisation described in Section 1.1.

6.9 Exclusion Criteria

A ticker is excluded from this signal if:

- Fewer than 30 trading days of price history are available (insufficient data for the rolling windows to stabilise).
- Any of $\%K_0$, $\%D_0$, or $\%K_{-1}$ is undefined (e.g. due to a zero-width trading range).

Excluded tickers receive the neutral default score of 50 in the composite.

7 Price Acceleration

7.1 Motivation

Price momentum captures the level of recent returns; price acceleration captures their *rate of change*. A stock transitioning from flat to strongly positive monthly returns is accelerating even if its cumulative return is still modest. This signal aims to identify stocks early in a move, before they appear on conventional momentum screens.

7.2 Definitions

Let $m = 21$ trading days represent one calendar month. Define the most recent monthly return:

$$R_{\text{recent}}(s) = \frac{C_0^s - C_{-m}^s}{C_{-m}^s} \tag{37}$$

Define six non-overlapping historical monthly returns for months $j = 1, 2, \ldots, 6$:

$$R_j(s) = \frac{C_{-jm}^s - C_{-(j+1)m}^s}{C_{-(j+1)m}^s} \tag{38}$$

and the average historical monthly return:

$$\bar{R}(s) = \frac{1}{J} \sum_{j=1}^{J} R_j(s), \qquad J = \min(6,\ \text{available months}) \tag{39}$$

The **discrete acceleration** is:

$$a_1(s) = R_{\text{recent}}(s) - \bar{R}(s) \tag{40}$$

7.3 EMA-Based Acceleration

Let $r_t = C_t/C_{t-1} - 1$ denote the daily return. Over the most recent $n = 126$ trading days, compute two exponential moving averages of daily returns:

$$E_{\text{short}} = \text{EMA}_{21}\big(\{r_t\}_{t=-125}^{0}\big), \qquad E_{\text{long}} = \text{EMA}_{126}\big(\{r_t\}_{t=-125}^{0}\big) \tag{41}$$

where EMA_λ uses span λ (i.e. decay factor $\alpha = 2/(\lambda + 1)$), and the subscript refers to the final (most recent) value of the series.

The **EMA acceleration** is:

$$a_2(s) = E_{\text{short}} - E_{\text{long}} \tag{42}$$

7.4 Raw Signal

The two acceleration measures are blended with equal weight. The EMA acceleration is rescaled by a factor of 100 to bring its magnitude (daily returns, order 10^{-3}) into approximate parity with the discrete acceleration (monthly returns, order 10^{-1}):

$$f_{\text{accel}}(s) = \frac{a_1(s) + 100\, a_2(s)}{2} \tag{43}$$

A ticker is excluded if fewer than 150 trading days of history are available or if $J < 3$.

8 News Attention

8.1 Motivation

In the small-cap universe, most stocks receive negligible media coverage. Any sustained or spiking attention is informative. The signal distinguishes between bullish attention (rising coverage on a rising stock, suggesting institutional interest) and bearish attention (rising coverage on a falling stock, suggesting negative developments).

8.2 Definitions

For each ticker, the following article counts are observed:

- $n_{30}(s)$ — total articles published in the past 30 calendar days
- $n_7(s)$ — total articles published in the past 7 calendar days

Let $n_{30}^{\max} = \max_{s \in \mathcal{U}} n_{30}(s)$.

8.2.1 Attention Level

The 30-day count is log-scaled to compress the heavily right-skewed distribution:

$$L(s) = \begin{cases} \dfrac{\ln(1 + n_{30}(s))}{\ln(1 + n_{30}^{\max})} & \text{if } n_{30}(s) > 0 \\ 0 & \text{otherwise} \end{cases} \tag{44}$$

8.2.2 Attention Surge

The expected weekly article rate is $\hat{w}(s) = n_{30}(s)/4.3$. The surge score measures deviation from this baseline:

$$\Gamma(s) = \begin{cases} \text{clip}\left(\dfrac{n_7(s)/\hat{w}(s) - 1}{2},\ -0.5,\ 1.0\right) & \text{if } \hat{w}(s) > 0 \\ 1.0 & \text{if } \hat{w}(s) = 0 \text{ and } n_7(s) > 0 \\ 0 & \text{otherwise} \end{cases} \tag{45}$$

A stock with no prior coverage that suddenly appears in the news receives the maximum surge score.

8.2.3 Price Direction

The 7-day price return is:

$$r_7(s) = \frac{C_0^s - C_{-5}^s}{C_{-5}^s} \tag{46}$$

(using 5 trading days $\approx$ 7 calendar days), with direction:

$$d(s) = \begin{cases} \operatorname{sgn}\big(r_7(s)\big) & \text{if } r_7(s) \neq 0 \\ 0.5 & \text{if } r_7(s) = 0 \end{cases} \tag{47}$$

8.3 Raw Signal

$$f_{\text{news}}(s) = \begin{cases} \big[0.5\, L(s) + 0.5\, \max\big(\Gamma(s),\, 0\big)\big] \cdot d(s) & \text{if } n_{30}(s) + n_7(s) > 0 \\ 0 & \text{otherwise} \end{cases} \tag{48}$$

The attention magnitude (always non-negative) is multiplied by the price direction to produce a signed signal. Stocks with zero news coverage receive a raw signal of exactly zero.

9 Relative Strength Index (RSI)

9.1 Motivation

RSI (Wilder, 1978) measures the speed and magnitude of recent price changes. Traditionally, readings above 70 are considered "overbought" in a mean-reversion framework. For momentum screening, the interpretation is inverted: RSI in the 60–80 range indicates a healthy uptrend, while extreme readings above 85 warrant caution as the move may be overextended.

9.2 Definition

Let $\delta_t = C_t^s - C_{t-1}^s$ denote the daily price change. Define:

$$G_t = \max(\delta_t,\ 0), \qquad \Lambda_t = \max(-\delta_t,\ 0) \tag{49}$$

The average gain and average loss are computed using Wilder's smoothing (an exponential moving average with $\alpha = 1/p$, where $p = 14$):

$$\bar{G}_t = \mathrm{EMA}_{1/p}\big(\{G_i\}\big)\Big|_t, \qquad \bar{\Lambda}_t = \mathrm{EMA}_{1/p}\big(\{\Lambda_i\}\big)\Big|_t \tag{50}$$

The relative strength and RSI are:

$$\mathrm{RS}_t = \frac{\bar{G}_t}{\bar{\Lambda}_t}, \qquad \mathrm{RSI}_t = 100 - \frac{100}{1 + \mathrm{RS}_t} \tag{51}$$

9.3 Momentum-Adjusted Transformation

The standard RSI $\in [0, 100]$ is transformed into a raw signal that rewards the momentum sweet spot and penalises extremes:

$$f_{\mathrm{RSI}}(s) = \begin{cases} \dfrac{\mathrm{RSI} - 50}{30} & \text{if } 50 \le \mathrm{RSI} \le 80 \\ \max\left(1 - \dfrac{\mathrm{RSI} - 80}{40},\ 0.2\right) & \text{if } \mathrm{RSI} > 80 \\ \max\left(\dfrac{\mathrm{RSI} - 50}{30},\ -1\right) & \text{if } \mathrm{RSI} < 50 \end{cases} \tag{52}$$

This yields the following profile:

RSI zone	Raw signal	Interpretation
RSI = 80	+1.00	Peak momentum (ceiling)
RSI = 65	+0.50	Strong momentum
RSI = 50	0.00	Neutral
RSI = 35	−0.50	Weak / bearish
RSI = 20	−1.00	Deeply bearish (floor)
RSI = 90	+0.75	Overextended (penalised from peak)
RSI = 100	+0.50	Maximally overextended

A ticker is excluded if fewer than 30 trading days are available.

Note: This signal carries a default weight of zero and is not included in the composite score. It remains available for custom weight configurations.

Appendix: Operator Definitions

For reference, the operators used throughout this document:

$$\mathrm{clip}(x,\ a,\ b) = \max\bigl(a,\ \min(x,\ b)\bigr) \tag{53}$$

$$\mathrm{sgn}(x) = \begin{cases} +1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases} \tag{54}$$

$$\mathrm{EMA}_{\lambda}\bigl(\{x_t\}\bigr)\Big|_T = \text{final value of exponential moving average with span } \lambda \tag{55}$$

$$\text{using decay } \alpha = 2/(\lambda+1),\ \text{evaluated at } t = T \tag{56}$$

$$\mathrm{EMA}_{1/p}\bigl(\{x_t\}\bigr)\Big|_T = \text{EMA with Wilder smoothing factor } \alpha = 1/p \tag{57}$$

$$\mathrm{percentile}(x \mid S) = \text{percentile rank of } x \text{ within set } S,\ \text{scaled } [0, 100] \tag{58}$$